2.6 Solving Absolute Value Inequalities

Name _____ Period ____ Date _____

Algebra I 2.6 · Student sheet

I can…

- solve absolute value inequalities of the form |ax + b| < c.
- solve absolute value inequalities of the form |ax + b| > c.
- use absolute value inequalities to solve real-life problems.

Vocabulary · fill in the missing word

Absolute value inequality — an inequality with a _____ inside absolute value bars

|x| < c — _____

|x| > c — _____

Part 1 · Distance thinking

A. Shade all numbers within 2 units of 5:

B. Shade all numbers more than 2 units from 5:

Graph	Absolute value inequality	Compound inequality
A		
B		

Part 2 · Notes and examples (fill in each note, then work the example)

Less than: |ax + b| < c → _____ < ax + b < _____ (“and”)

Example 1

Solve |x + 4| < 6 and graph.

Greater than: |ax + b| > c → ax + b < _____ or ax + b > _____ (“or”)

Example 2

Solve |2x − 1| ≥ 9 and graph.

Isolate first: Isolate the absolute value _____.

Example 3

Solve $3|x - 2| + 1 \leq 13$.

Example 4

A thermostat is set to 70°F. The temperature stays within 3°F of the setting. Write and solve an absolute value inequality.

Part 3 · Partner practice

Partner 1

Solve $|x - 3| > 5$.

Partner 2

Solve $|x| + 4 < 2$.